XAI3 Deliverable: Partial Dependency Plot

Model-agnostic methods applied to Random Forest

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1. Introduction and methodology

1.1. Objective

Apply the **Partial Dependency Plot (PDP)** method to two regression problems solved using *Random Forest*, in order to **interpret the behavior learned by the model** without accessing its internal structure (*model-agnostic* approach).

1.2. Fundamentals of PDP

Let $\hat{f}$ be a trained model and $\mathbf{x} = (x_S, x_C)$ be the input vector, where x_S is the subset of variables of interest and x_C are the remaining variables. The PDP is defined as:

$$\hat{f}_{x_S}(x_S) = \mathbb{E}_{x_C}[\hat{f}(x_S, x_C)] \approx \frac{1}{n}\sum_{i=1}^{n} \hat{f}(x_S, x_C^{(i)})$$

That is: for each value of x_S, the model's prediction is averaged while holding the other variables fixed at their observed values.

Key limitation. The PDP assumes independence between x_S and x_C; when strong correlations exist, the averaged values may correspond to unrealistic combinations of variables. For this reason, the graphs are accompanied by the **empirical distribution** of the variables (bottom `rug`).

1.3. Package load

```
library(randomForest)
library(ggplot2)
library(dplyr)
library(gridExtra)
library(viridis)
```

1.4. Generic PDP Function

The algorithm is implemented **manually** below (instead of using the `pdp` package) to clearly show the calculation:

```
# One-dimensional PDP.
# For each grid value, the feature is replaced across the entire dataset #
and the model predictions are averaged.
compute_pdp <- function(model, data, feature, n_points = 50) {
  values <- seq(min(data[[feature]]), max(data[[feature]]),
                length.out = n_points)
  yhat <- vapply(values, function(v) {
    d <- data
```

```
    d[[feature]]  <-  v
    mean(predict(model,   d))
  },  numeric(1))
  data.frame(x = values, yhat = yhat)
}

# Two-dimensional PDP. Same idea applied to two simultaneous features.
compute_pdp_2d <- function(model, data, f1, f2, n_points = 20) {
  v1    <-     seq(min(data[[f1]]),     max(data[[f1]]),     length.out     =
  n_points)  v2  <-  seq(min(data[[f2]]),  max(data[[f2]]),  length.out
  = n_points) grid <- expand.grid(v1 = v1, v2 = v2)
  names(grid)  <-  c(f1,  f2)
  grid$yhat   <-   vapply(seq_len(nrow(grid)),   function(i)
    { d <- data
    d[[f1]] <- grid[[f1]][i]
    d[[f2]]                    <-
    grid[[f2]][i]
    mean(predict(model,   d))
  },
  numeric(1))
  grid
}
```

2. Exercise 1 — One-dimensional PDP: Bike Rental

2.1. Loading and preparing data

```
day <- read.csv("data/day.csv")

# The 'instant' column is a daily sequential index starting at 1 # and
corresponding to January 1, 2011. Therefore, it represents exactly # "days since
2011".
day$days_since_2011  <-  day$instant

# Variables to be included in the Random Forest. This includes both
# those to be analyzed using PDP and the rest of the relevant variables, # so that
the model has the full context.
bike_features <- c("days_since_2011", "temp", "hum", "windspeed",
                   "season", "yr", "mnth", "holiday", "weekday",
                   "workingday", "weathersit")
bike_data <- day[, c(bike_features, "cnt")]

cat("Dimensions of the dataset:", nrow(bike_data), "x", ncol(bike_data), "\n")
```

```
## Dataset dimensions: 731 x 12
```

> **Note on the scale of the variables.** In the original *Bike Sharing Dataset*, `temp`, `hum`, and `windspeed` are **normalized to the range [0, 1]**: `temp` divided by 41 °C, `hum` by 100%, and `windspeed` by 67. This normalization is maintained as-is in the plots to preserve consistency with the dataset.

2.2. Training the Random Forest

```
set.seed(42)
rf_bike <- randomForest(cnt ~ ., data = bike_data,
                        ntree = 500, importance = TRUE)

r2_bike <- 1 - rf_bike$mse[length(rf_bike$mse)] / var(bike_data$cnt)
cat("R² Out-Of-Bag:", round(r2_bike, 3), "\n")
```

```
## R² Out-Of-Bag: 0.892
```

The model achieves a very high fit (R^2 OOB ≈ 0.892), which supports the reliability of the PDPs calculated below.

2.3. Calculation of the PDPs

```
pdp_days <- compute_pdp(rf_bike, bike_data, "days_since_2011")
pdp_temp <- compute_pdp(rf_bike, bike_data, "temp")
pdp_hum <- compute_pdp(rf_bike, bike_data, "hum")
pdp_wind <- compute_pdp(rf_bike, bike_data,
"windspeed")
```

2.4. Visualization

```
plot_pdp <- function(pdp_df, rug, xlab, title) {
  ggplot(pdp_df, aes(x = x, y = yhat)) +
    geom_line(color = "#2166ac", linewidth = 1.1) +
    geom_rug(data = data.frame(x = rug), aes(x = x),
             sides = "b", alpha = 0.25,
             length = unit(0.025, "npc"),
             inherit.aes = FALSE) +
    labs(title = title, x = xlab, y = "Predicted count") +
    theme_bw(base_size = 11) +
    theme(plot.title = element_text(face = "bold", size = 11))
}

p1 <- plot_pdp(pdp_days, bike_data$days_since_2011,
               "Days since 2011", "Days since 2011")
p2 <- plot_pdp(pdp_temp, bike_data$temp,
               "Temperature (norm.)", "Temperature")
p3 <- plot_pdp(pdp_hum, bike_data$hum,
               "Humidity (norm.)",
"Humidity") p4 <- plot_pdp(pdp_wind,
bike_data$windspeed,
               "Wind Speed (norm.)", "Wind")

grid.arrange(p1, p2, p3, p4, ncol = 2)
```

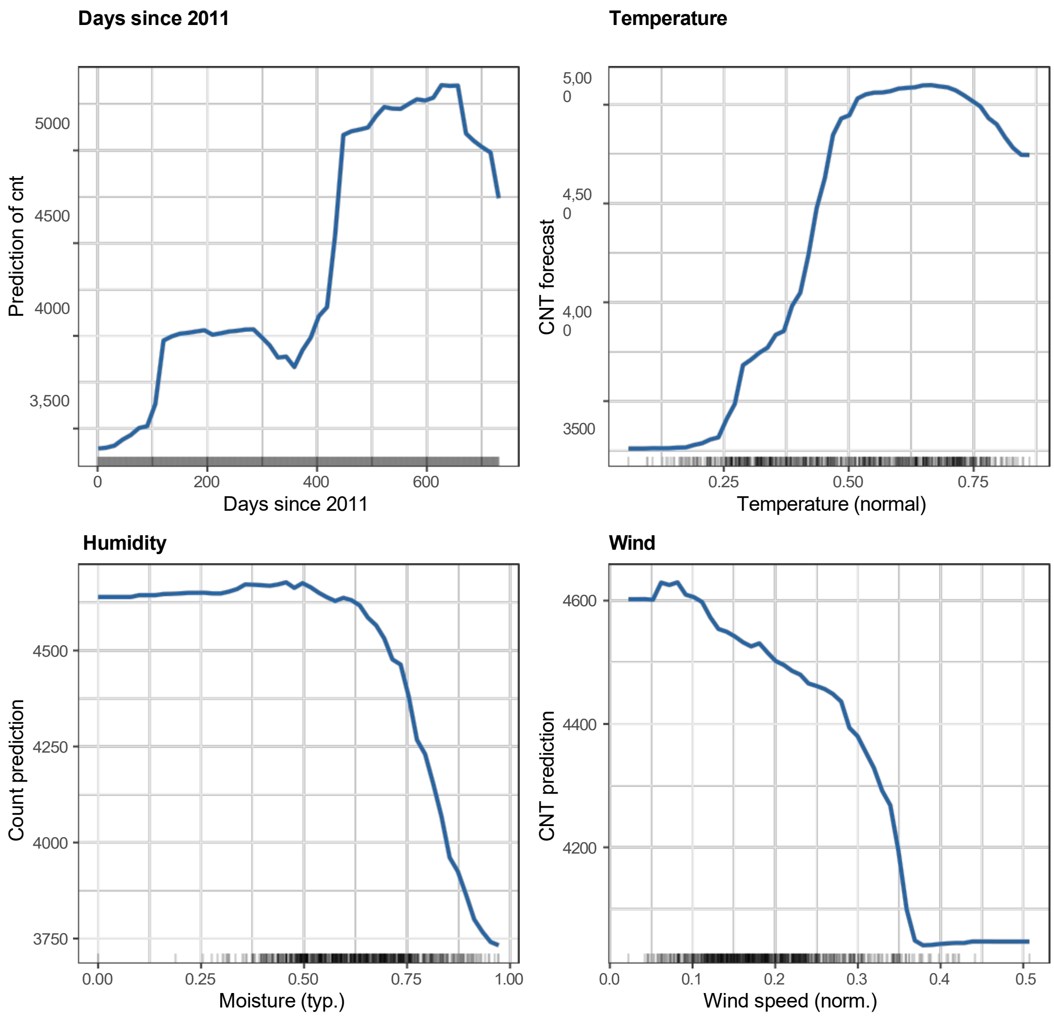

Figure 1: One-dimensional PDPs for the bike-sharing model. The blue line shows the marginal effect learned by the Random Forest; the lower plot represents the empirical distribution of the corresponding variable.

2.5. Analysis and Interpretation

Days since 2011. A **clear and practically monotonic upward trend** is observed: the predicted number of rentals rises from approximately 3,000 in early 2011 to approximately 5,500 by the end of 2012. This pattern reflects the **gradual adoption of the bike-sharing service** over those two years, a purely temporal effect independent of weather. The slope flattens toward the end, suggesting **partial saturation** of growth.

Temperature. There is a **positive, nonlinear** relationship **with an asymmetric bell curve**: starting at approximately 0.15 (≈ 6 °C), rentals grow rapidly, **peak around 0.65–0.70** (≈ 27–29 °C), and decline slightly at extreme values. This aligns with intuition: temperatures that are too cold or excessively hot discourage bicycle use. The curve confirms that the model has captured a **thermal optimum** rather than a purely linear relationship.

Humidity. The relationship is **negative**: the prediction remains relatively flat up to humidity values close to 0.6 and then **decreases sharply**. High humidity is often associated with rainy or cloudy days, conditions that deter users. The empirical density is concentrated in the 0.4–0.8 range, where the effect is most reliable.

Wind speed. Negative and approximately monotonic effect: the stronger the wind, the fewer users. The most pronounced drop occurs in the 0.1–0.4 range. For higher values, the effect diminishes, partly because the density of observations is very low (sparse coverage) and the model extrapolates with less confidence.

3. Exercise 2 — Two-dimensional PDP: humidity and temperature

3.1. Random sample

Calculating the 2D PDP requires evaluating the model over $|grid_1| \times |grid_2| \times n$ rows. To keep the computational cost reasonable, a **random sample** is drawn from the dataset, as recommended in the problem statement:

```r
set.seed(42)
bike_sample <- bike_data[sample(nrow(bike_data), 300), ]
cat("Sample size:", nrow(bike_sample), "rows\n")
```

```
## Sample size: 300 rows
```

3.2. Calculating the 2D PDP

```r
pdp_2d <- compute_pdp_2d(rf_bike, bike_sample,
                         f1 = "hum", f2 = "temp",
                         n_points = 25)
```

3.3. Visualization

To avoid gaps between cells, `width` and `height` are calculated based on the *grid* step. The empirical density of `hum` and `temp` is overlaid using `geom_rug()` in white on the bottom and left axes, indicating the regions where the PDP is most reliable.

```r
tile_w <- diff(range(pdp_2d$hum)) / 24
tile_h <- diff(range(pdp_2d$temp)) / 24

ggplot(pdp_2d, aes(x = hum, y = temp, fill = yhat)) +
  geom_tile(width = tile_w, height = tile_h) +
  geom_rug(data = bike_sample, aes(x = hum, y = temp),
           inherit.aes = FALSE,
            alpha = 0.5, color = "white",
           length = unit(0.02, "npc")) +
   scale_fill_viridis_c(option = "plasma",
                        name = "predicted_cnt") +
  labs(title = "2D PDP: Humidity x Temperature",
       x = "Humidity (norm.)",
       y = "Temperature (norm.)") +
  theme_bw(base_size = 11) +
  theme(plot.title = element_text(face = "bold"),
        legend.position = "right")
```

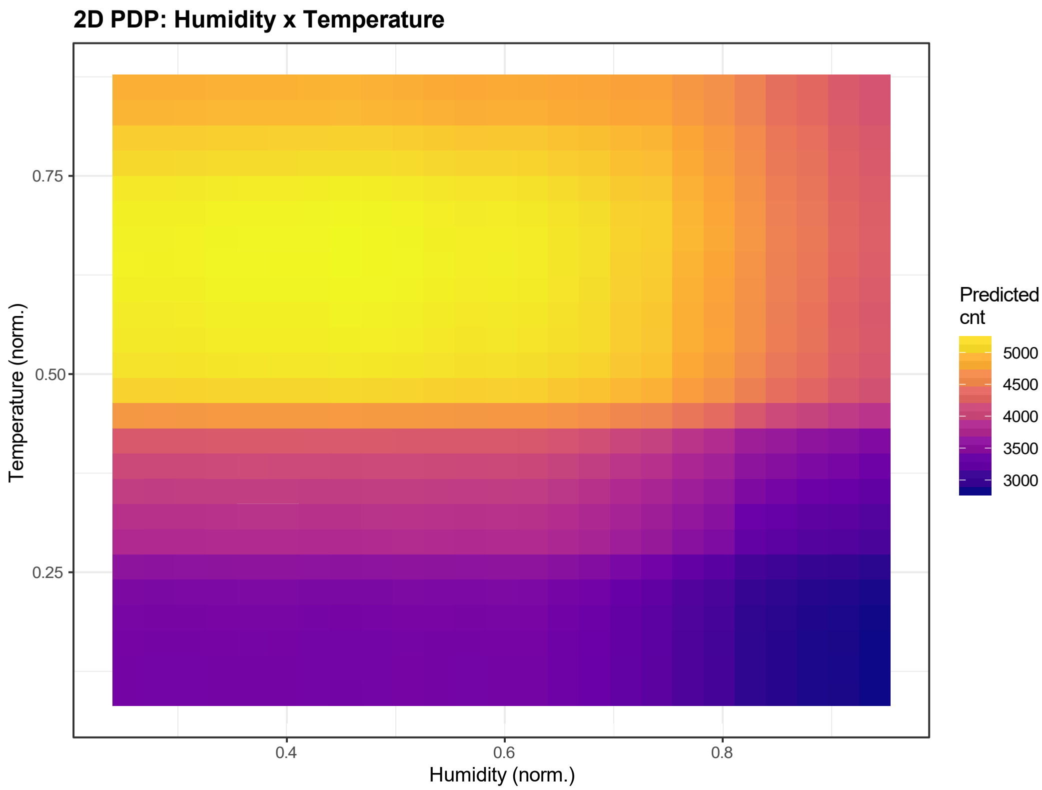

Figure 2: Two-dimensional PDP for the humidity-temperature interaction. The color represents the Random Forest prediction for the number of rentals. The white marks on the axes show the empirical density of each variable.

3.4. Distribution of the variables

As a complement to the PDP, the joint marginal distribution of the two variables is shown. This allows us to identify areas of the plot where the model extrapolates due to a lack of data.

```r
ggplot(bike_data, aes(x = hum, y = temp)) +
  geom_point(alpha = 0.4, color = "#2166ac", size = 1.4) +
  geom_density_2d(color = "#b2182b", linewidth = 0.6) +
  labs(title  =  "Joint  humidity-temperature  distribution",
       x = "Humidity (norm.)",
       y = "Temperature (norm.)") +
  theme_bw(base_size = 11) +
  theme(plot.title = element_text(face = "bold"))
```

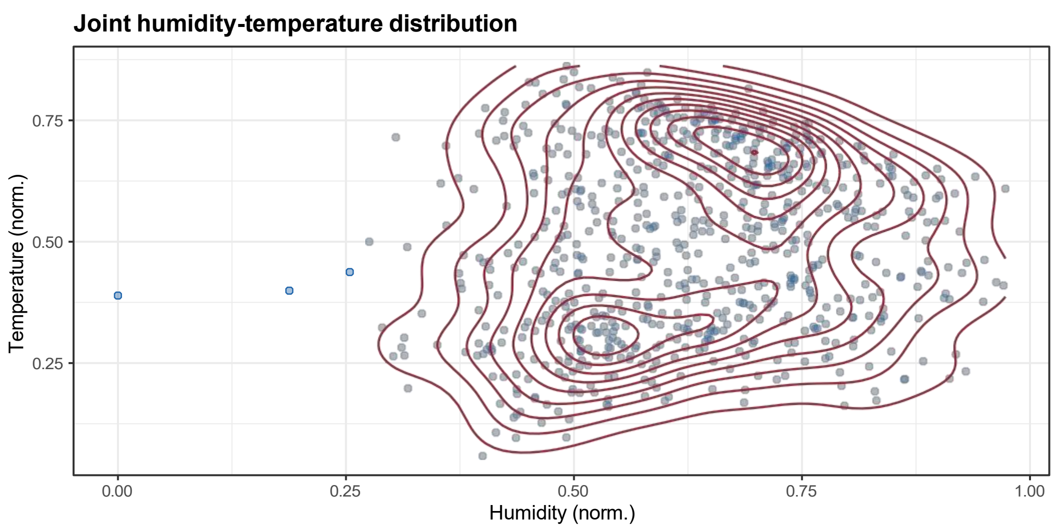

Figure 3: Joint empirical distribution of humidity and temperature in the dataset (points) with superimposed 2D density curves.

3.5. Analysis and Interpretation

The two-dimensional plot confirms and enriches the conclusions obtained in the 1D PDPs:

- **The highest prediction** (bright yellow zone) is found in the region of **high temperature ($\sim$ 0.6–0.8) and low-to-moderate humidity ($\sim$ 0.3–0.6)**: warm but dry days, optimal conditions for cycling.
- The **lowest prediction** (dark zone) corresponds to the combination of **low temperature and high humidity**, that is, cold and humid days, typically wintry and unpleasant.
- The color transition is **diagonal**, suggesting a **smooth interaction** between the two variables: the positive effect of temperature is **mitigated by high humidity**, and the negative effect of humidity is partially offset by favorable temperatures.
- In **areas with low empirical density** (upper left corner: very high temperature and very low humidity; lower right corner: very low temperature and very high humidity), PDP predictions should be interpreted with caution, as the model extrapolates on combinations that are poorly represented in the data. This is precisely the limitation warned by PDP theory when there is correlation between variables.

4. Exercise 3 — PDP: Housing Price Prediction

4.1. Loading and Random Sampling

The kc_house_data.csv dataset contains approximately 21,000 homes. To make the PDP calculation manageable, we work with a **random sample of 1,000 observations**, both for training the model and for evaluating the PDP:

```
house <- read.csv("data/kc_house_data.csv")
cat("Original dataset:", nrow(house), "rows\n")
```

```
## Original dataset: 21,613 rows
```

```
set.seed(42)
house_sample <- house[sample(nrow(house), 1000), ]

# The 6 features specified in the problem statement.
house_features <- c("bedrooms", "bathrooms", "sqft_living",
                    "sqft_lot", "floors", "yr_built")
house_data <- house_sample[, c(house_features, "price")]

cat("Sample used:", nrow(house_data), "rows\n")
```

```
## Sample used: 1000 rows
```

```
summary(house_data$price)
```

```
##     Min. 1st Quartile  Median
      Mean 3rd Quartile        Max. ##
      89,950 314,750 449,975 531,884 627,000
3,300,000
```

4.2. Random Forest Training

```
set.seed(42)
rf_house <- randomForest(price ~ ., data = house_data,
                         ntree = 500, importance = TRUE)

r2_house <- 1 - rf_house$mse[length(rf_house$mse)] / var(house_data$price)
cat("Out-Of-Bag R²:", round(r2_house, 3), "\n")
```

```
## R² Out-Of-Bag: 0.563
```

The model explains approximately 56.3% of the price variance on unseen data (*out-of-bag* validation), a reasonable result considering that only 6 variables out of the total available are used and the sample size is small.

4.3. Calculation of the PDPs

```r
pdp_bedrooms <- compute_pdp(rf_house, house_data, "bedrooms")
pdp_bathrooms <- compute_pdp(rf_house, house_data, "bathrooms")
pdp_sqftliv <- compute_pdp(rf_house, house_data, "sqft_living")
pdp_floors     <- compute_pdp(rf_house, house_data, "floors")
```

4.4. Visualization

```r
plot_pdp_house <- function(pdp_df, rug, xlab, title) {
  ggplot(pdp_df, aes(x = x, y = yhat)) +
    geom_line(color = "#b2182b", linewidth = 1.1) +
    geom_rug(data = data.frame(x = rug), aes(x = x),
             sides = "b", alpha = 0.25,
             length = unit(0.025, "npc"),
             inherit.aes = FALSE) +
    scale_y_continuous(labels = function(v) paste0("$", format(v / 1000, big.mark = ",",    trim = TRUE),
    labs(title = title, x = xlab, y = "Predicted Price") +
    theme_bw(base_size = 11) +
    theme(plot.title = element_text(face = "bold", size = 11))
}

q1 <- plot_pdp_house(pdp_bedrooms, house_data$bedrooms,
                     "Bedrooms", "Bedrooms")
q2 <- plot_pdp_house(pdp_bathrooms, house_data$bathrooms,
                     "Bathrooms", "Baños")
q3 <- plot_pdp_house(pdp_sqftliv, house_data$sqft_living,
                     "Sqft living", "Living area (sqft)") q4 <-
plot_pdp_house(pdp_floors,        house_data$floors,
                     "Floors", "Floors")

grid.arrange(q1, q2, q3, q4, ncol = 2)
```

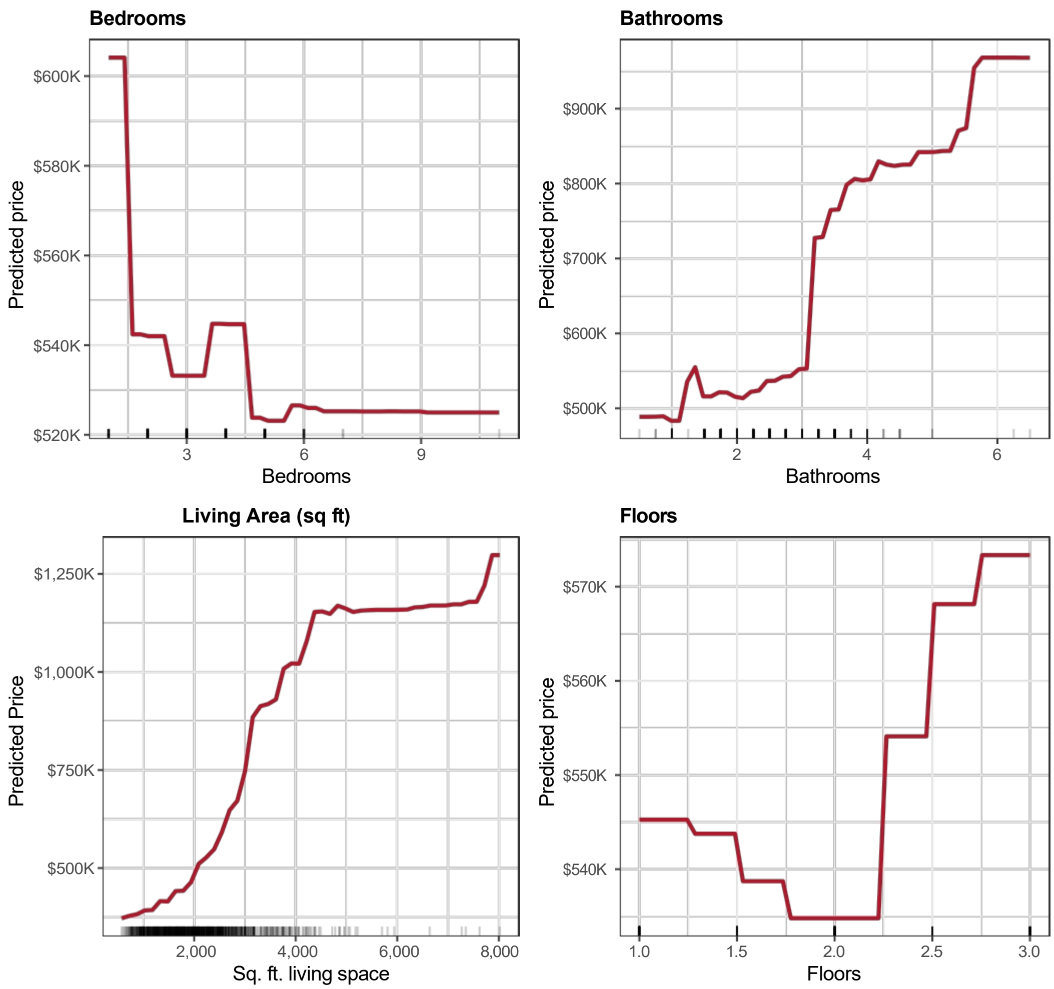

Figure 4: One-dimensional PDPs for price prediction. The vertical axis is expressed in U.S. dollars.

4.5. Analysis and Interpretation

Bedrooms. The effect is **positive but clearly diminishing**: adding bedrooms increases the price only up to a certain point (3–5 bedrooms), where the curve stabilizes or even declines slightly. This is consistent with the real estate market: for the same floor area, too many bedrooms mean smaller rooms, which **ceases to add value**. Furthermore, the empirical density is concentrated in the 2–5 bedroom range, so the extremes of the axis are less reliable.

Bathrooms. A **positive and approximately monotonic** relationship, more pronounced than that of bedrooms. Each additional bathroom contributes to a sustained increase in price, especially in the 1.5–4 range. Bathrooms are a clear indicator of a home's **quality and comfort**, which justifies this behavior.

Living area (sqft_living). This is the variable with **the greatest influence**, exhibiting a **positive, strong, and nearly linear** relationship in the central range. The slope flattens at very high values (> 6,000 sqft), where the market becomes highly specific and the model has few observations. This result aligns with the economic intuition that price is largely determined by the size of the home.

Floors. Positive but modest effect: moving from one to two floors increases the price appreciably, but beyond that, the gain is small. It should be noted that the variable takes only a few discrete values (1, 1.5, 2, 2.5, 3, 3.5), so the PDP between those values is the result of interpolation rather than actual model behavior.

5. General conclusions

Throughout this study, the **Partial Dependency Plot** method was applied to two *Random Forest* models, demonstrating its usefulness as a *model-agnostic* XAI tool:

1. **1D Interpretation (bicycles).** The PDPs revealed clear and consistent patterns: sustained growth in demand over time, a thermal optimum around 28 °C, and deterrent effects of humidity and wind. This demonstrates that the Random Forest captures nonlinear structure without the need to define manual transformations.
2. **2D interaction (bicycles).** The two-dimensional `humidity × temperature` PDP revealed a **multiplicative, non-additive effect**: the most favorable combination (high temperature + low humidity) boosts the prediction, while the opposite combination strongly depresses it. The joint empirical distribution also highlighted **areas where the PDP should be interpreted with caution**, as the model extrapolates beyond the actual support.
3. **1D Interpretation (Housing).** The price analysis showed that `sqft_living` is the dominant variable, while `bedrooms` exhibit **diminishing returns** and `floors` have a modest effect. These results align with economic intuition regarding the real estate sector and reinforce confidence in the model.

Recognized limitations. The PDP represents the **average** effect: it does not capture heterogeneity across instances (individual effects) nor does it distinguish interactions not included in the subset x_S . To complement the analysis, it would be reasonable to use tools such as **ICE plots**, **ALE plots** (which mitigate the problem of dependence between variables), or **SHAP values**.

6. Session Information

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products:
default ##LAPACK version
3.12.1 ##
## locale:
## [1] LC_COLLATE=Spanish_Spain.utf8 LC_CTYPE=Spanish_Spain.utf8
## [3] LC_MONETARY=Spanish_Spain.utf8 LC_NUMERIC=C
## [5] LC_TIME=Spanish_Spain.utf8
##
## time zone:
Europe/Madrid ## tzcode
source: internal ##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] viridis_0.6.5       viridisLite_0.4.2   gridExtra_2.3
## [4] dplyr_1.1.4         ggplot2_4.0.1       randomForest_4.7-1.2
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6       compiler_4.5.2     tidyselect_1.2.1   scales_1.4.0
```

```
## [5]  yaml_2.3.12         fastmap_1.2.0     R6_2.6.1           labeling_0.4.3
## [9]  generics_0.1.4      isoband_0.3.0     knitr_1.51         MASS_7.3-65
## [13] tibble_3.3.0        pillar_1.11.1     RColorBrewer_1.1-3 rlang_1.1.6
## [17] xfun_0.55           S7_0.2.1          otel_0.2.0         cli_3.6.5
## [21] withr_3.0.2         magrittr_2.0.4    digest_0.6.39      grid_4.5.2
## [25] lifecycle_1.0.5     vctrs_0.6.5       evaluate_1.0.5     glue_1.8.0
## [29] farver_2.1.2        codetools_0.2-20  rmarkdown_2.30     tools_4.5.2
## [33] pkgconfig_2.0.3     htmltools_0.5.9
```